

ITI - AMS Layout Course - Lab 02

Logic Gate Layout, Layers, GDS, and Fabrication

Outline

- Part1: Design and verification of an inverter using LVT devices
- Part2: Analyze the layers using layer palette
- Part3: Exporting, importing and updating GDS using layer mapping

Objective

1. To be familiar with the PDK technology layers
2. To be able to design and verify logic circuits
3. To relate between the fabrication process and the used devices in the layout

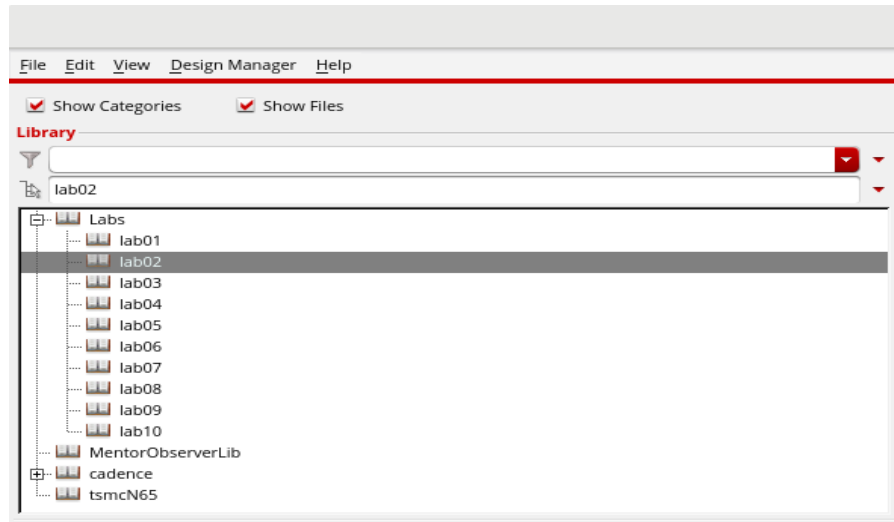
Tasks

1. Report the progress done in part 1
2. Linking between the layers used in part 1 and the fabrication steps
3. Replace LVT devices with HVT devices using layers from layer palette

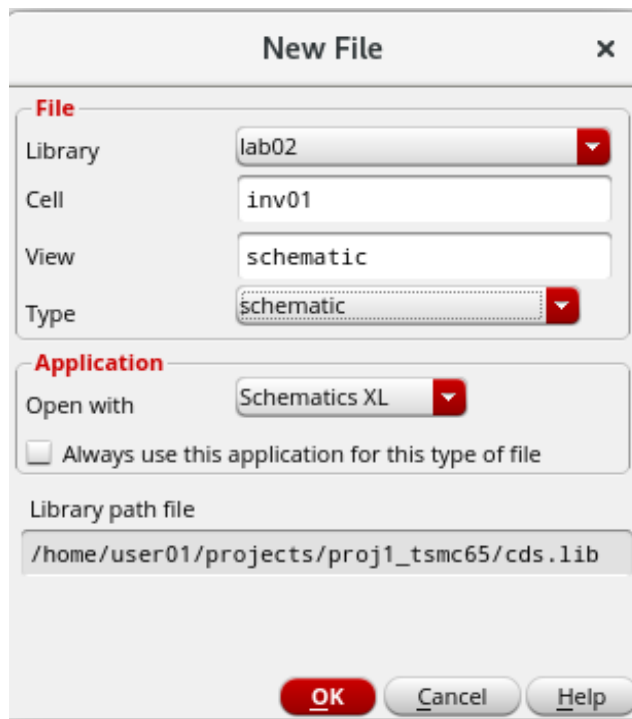
Important Note: Always make sure to save the machine's state before closing it. This is to avoid losing any work which may be used afterwards in other labs and/or projects.

Part1: Design and verification of an inverter using LVT devices

1. Create schematic view for the design
 - a. From “Labs” library select “lab02”



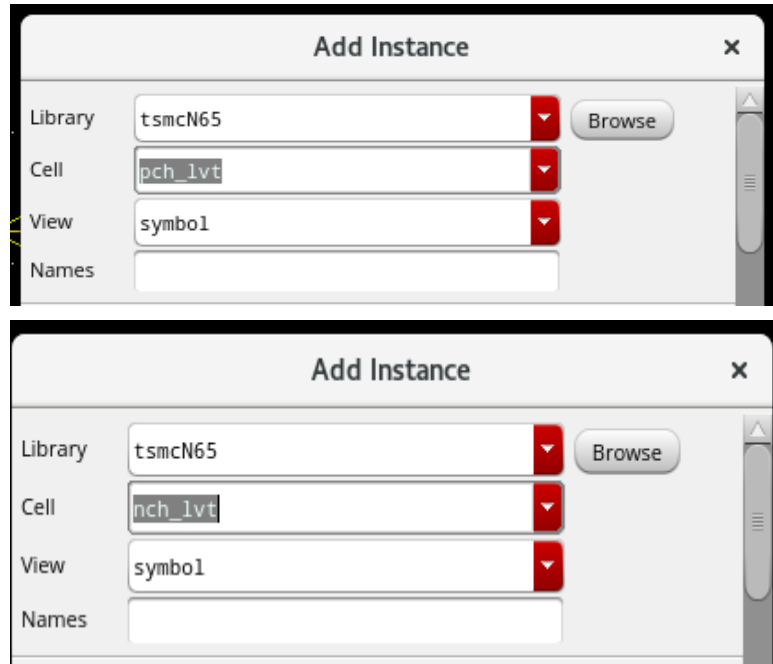
- b. Create a new schematic view with name “inv01”



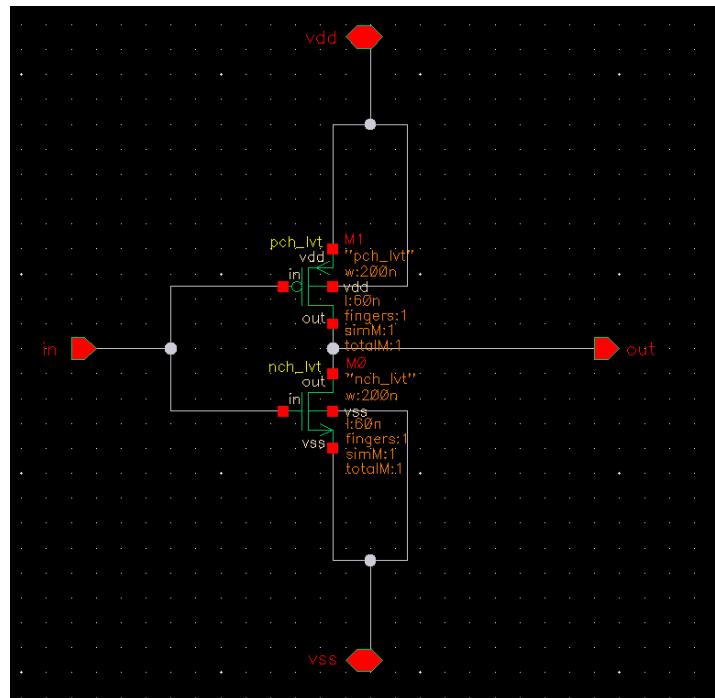
- c. After opening schematic press “i” to instantiate instance

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- i. From tsmcN65 select “pch_lvt” & “nch_lvt” devices with any sizing

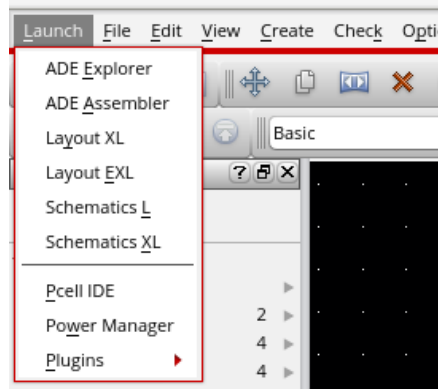


- d. Connect devices as shown below



2. Creating layout view

- From toolbar select launch > Layout XL

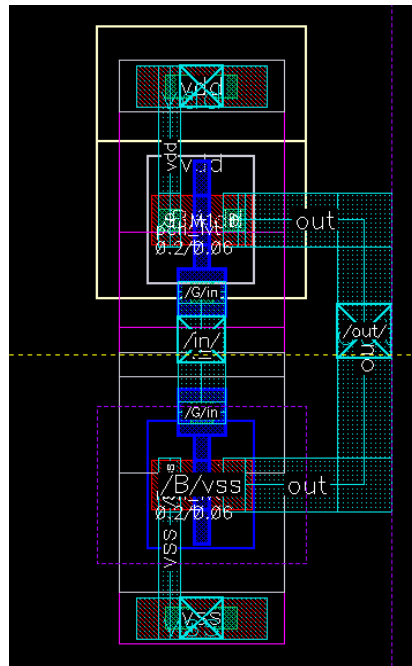


- Click on the “Generate all from source” option from the lower toolbar, place devices and pins, then continue the placement and routing of the devices.



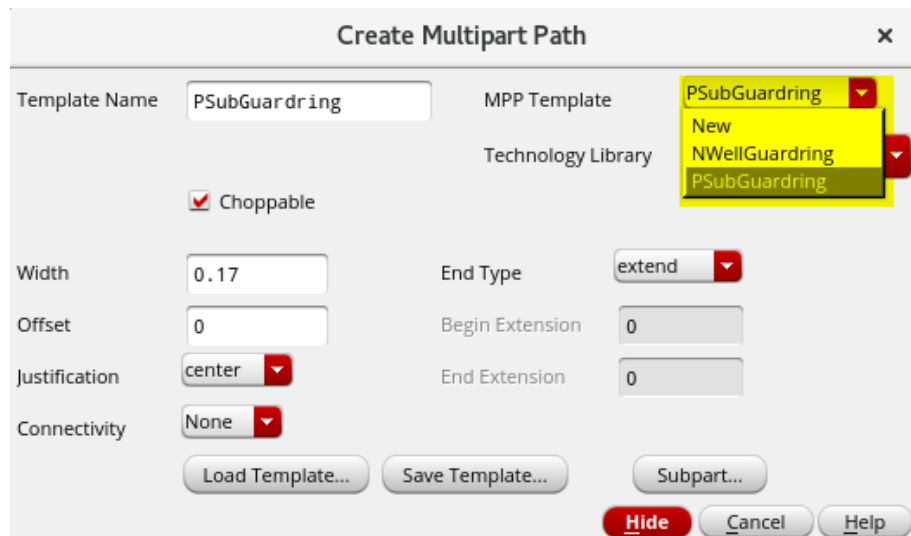
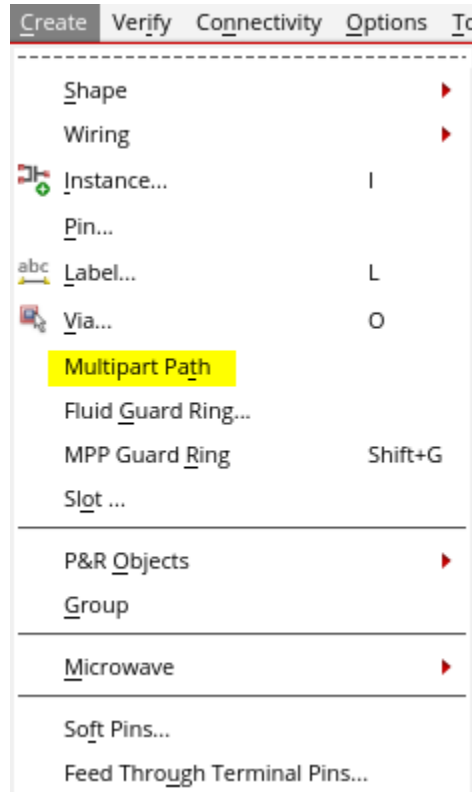
- Connect devices as in schematic and they should look like the below layout at the end, however try to optimize it more in area.

Note: Logic cells are used in large counts, so it is better to design them as tight as possible in area., try to make it optimized as much



- Placing straps:

- i. To connect the bulk of the device and prevent latch-up, we have to use guard rings or substrate straps
- ii. To save area we will add straps that consist of (M1, PP) for the NMOS device and (M1, NP, and NW) for the PMOS device
- iii. To place psub or nwell taps, goto “Create” > “Multipart Path” then press “F3” to show the feature options and select the proper tap type

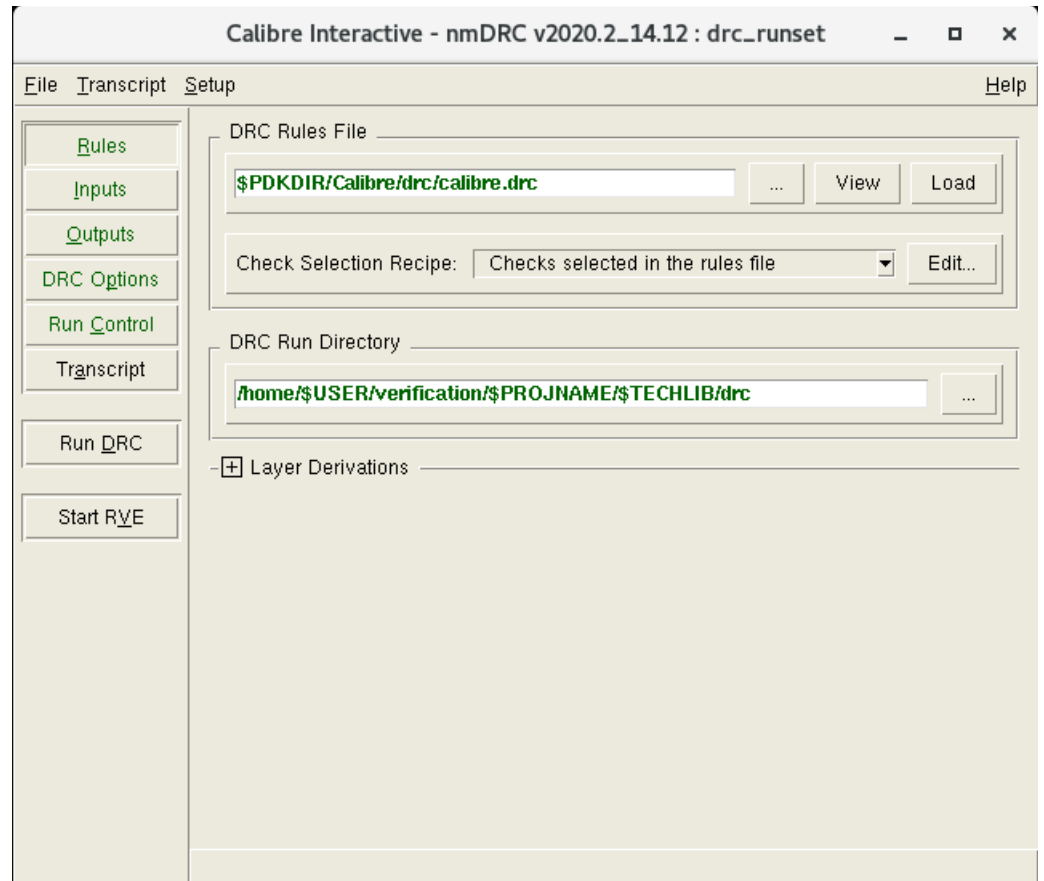


To draw the strap/tap, click on a point in the layout then move with the cursor to another point and press “Enter”

3. Start the verification of your design

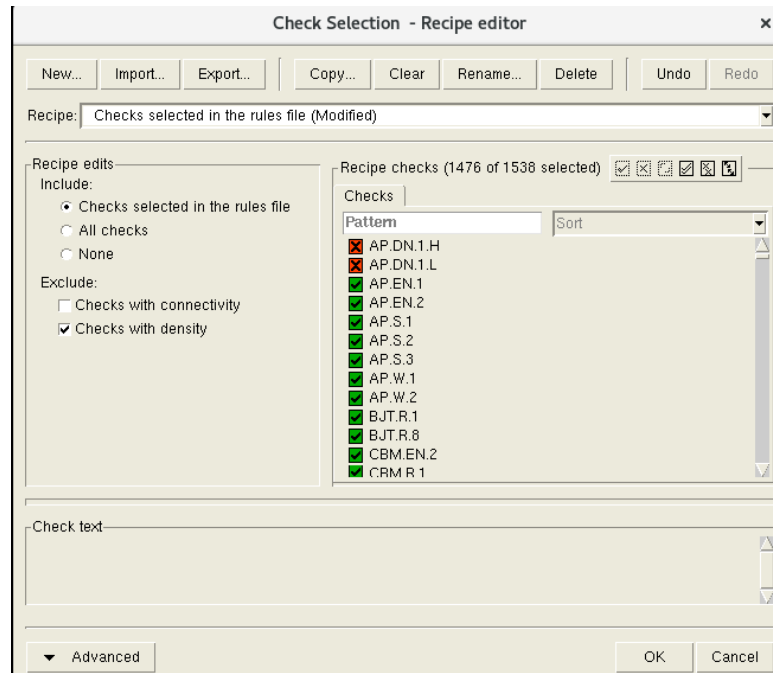
a. DRC : From layout view upper toolbar choose Calibre > Run DRC

i. Make sure that selected rule file is : \$PDKDIR/Calibre/drc/calibre.drc

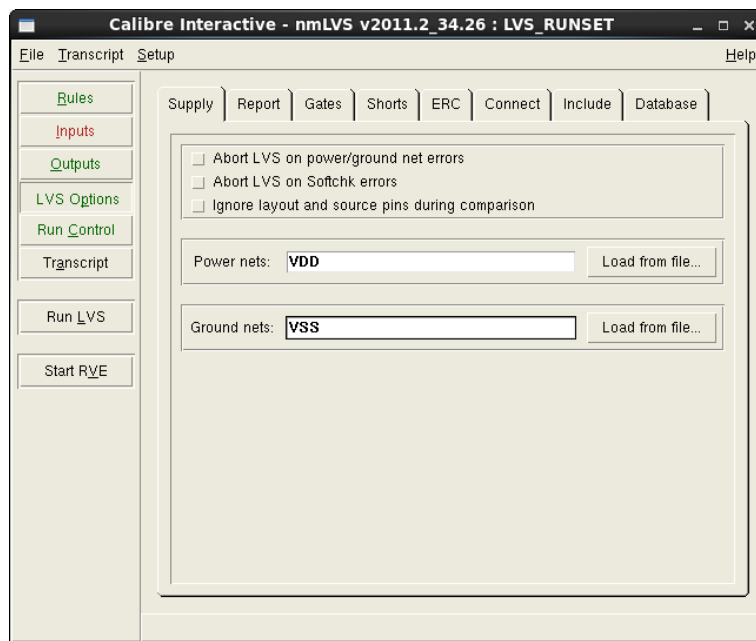


ii. From rules > edit, enable exclude “checks with density” to filter the density drc errors.
Note: This option for this lab only but on the higher levels no rules are excluded

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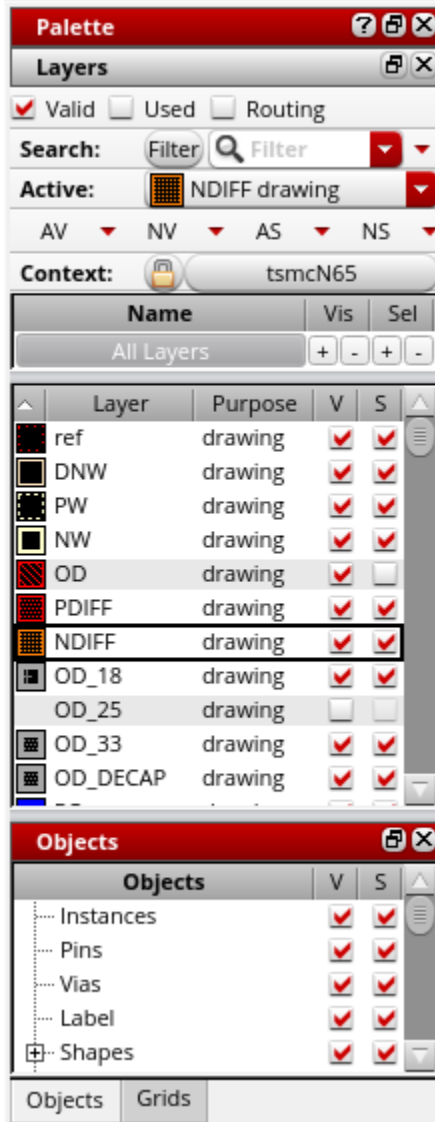


- B. LVS : Apply the first step similar to DRC but choose RUN LVS
- Enter the supply names then RUN LVS



Part2: Analyze the layers being used in the design of Part 1 using the “Layers Palette”

1. Getting familiar with the “Palette”



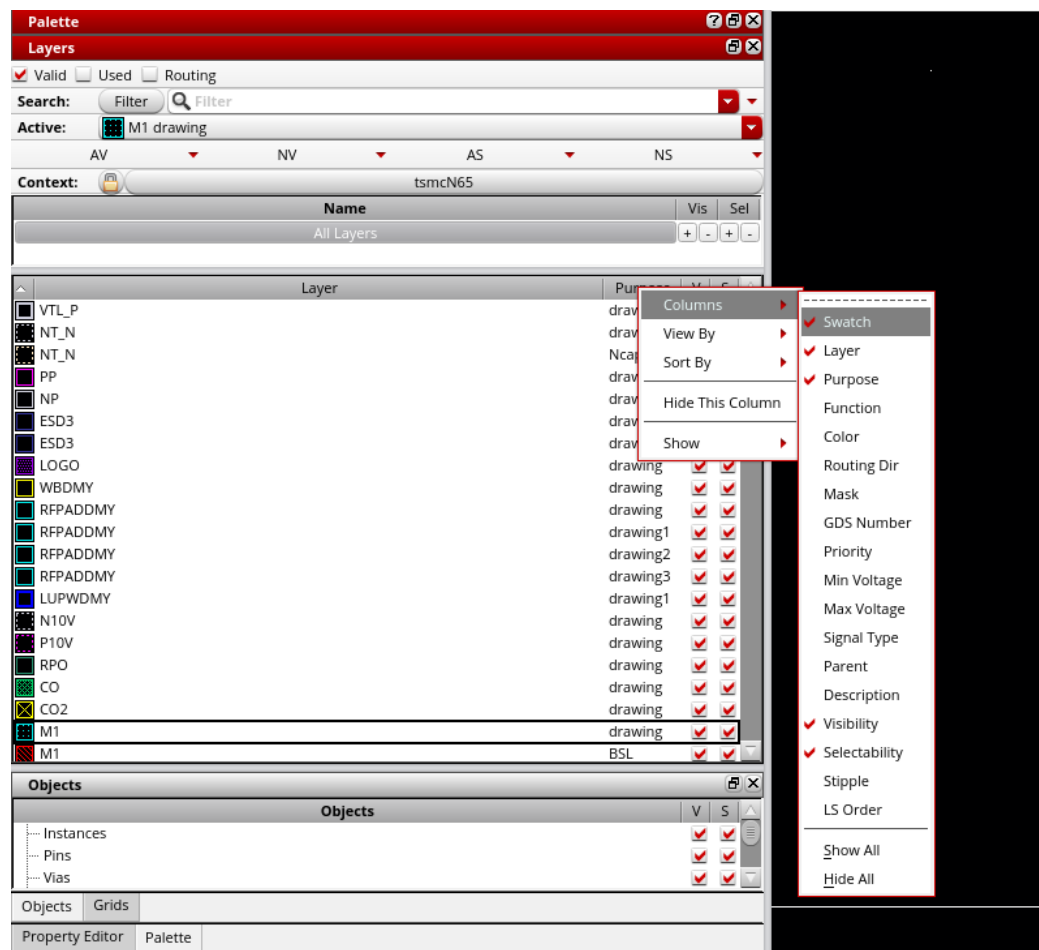
- a. The “Palette” in Virtuoso is a graphical user interface (GUI) that allows you to view and select the layers and objects that are used in your layout. The Palette is divided into two panes; the layer list and the Objects pane. The layer list pane shows a list of all the layers that are defined in the current library. You can use the layer list pane to control the selectivity and visibility of the layers being used in the layout. While the Objects pane enables you to control the selectivity and visibility of the objects being used in the layout.

The layer palette is a powerful tool that can help you to visualize and manage the layers in your layout. You can use the layer palette to:

- View the layers that are used in your layout
- Select the layers that you want to view or edit
- Change the properties of the selected layer
- Create and save layer set

b. Here are some of the features of the layer palette

- You can use the filter field to filter the list of layers.
- You can use the sort button to sort the list of layers by name, number, or type.
- You can use the view options button to customize the appearance of the layer palette.
- You can use the Layer Sets button to create and save layer sets.



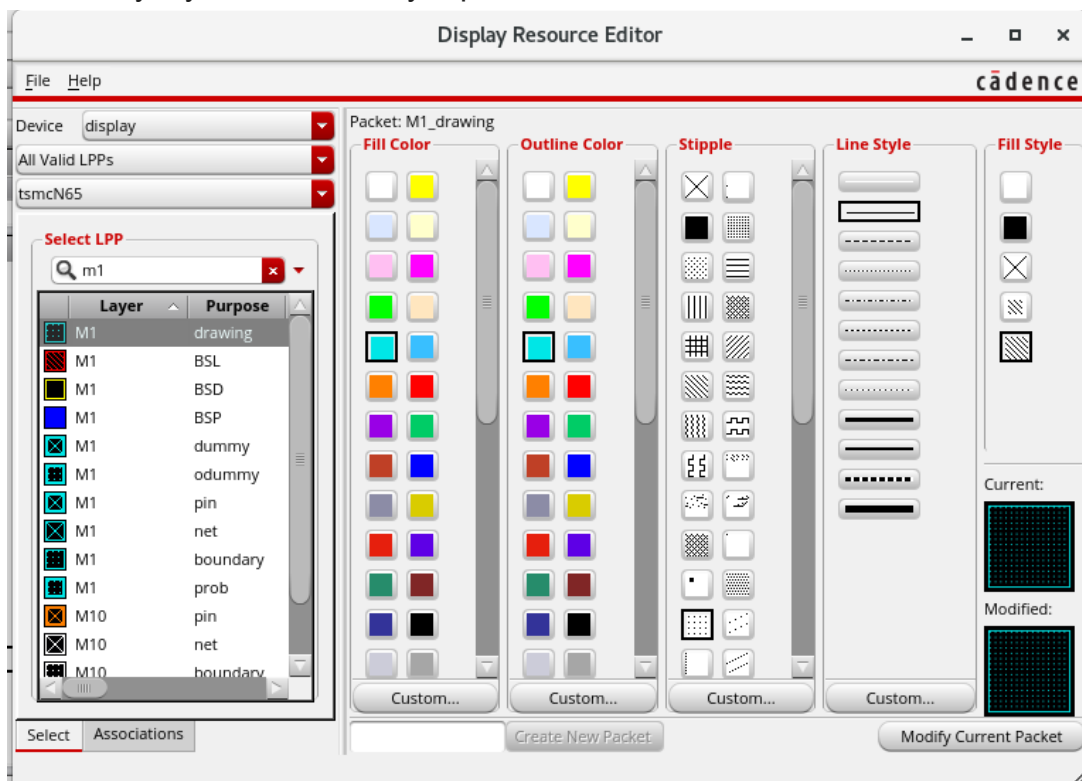
2. Display file

- a. The “display.drf” file is a text file that contains the display properties for layers in a Cadence Virtuoso layout. The display properties include the line style, fill style, fill and line colors, and other properties.

The display.drf file is used by Virtuoso to determine how to display the layers in a layout. When you open a layout in Virtuoso, the display.drf file for the library that the layout is associated with is loaded. The display properties in the display.drf file are then used to display the layers in the layout. You can use the Display Resource Editor to view and edit the display properties in a display.drf file.

The Display Resource Editor is a graphical user interface (GUI) that allows you to easily change the display properties of layers.

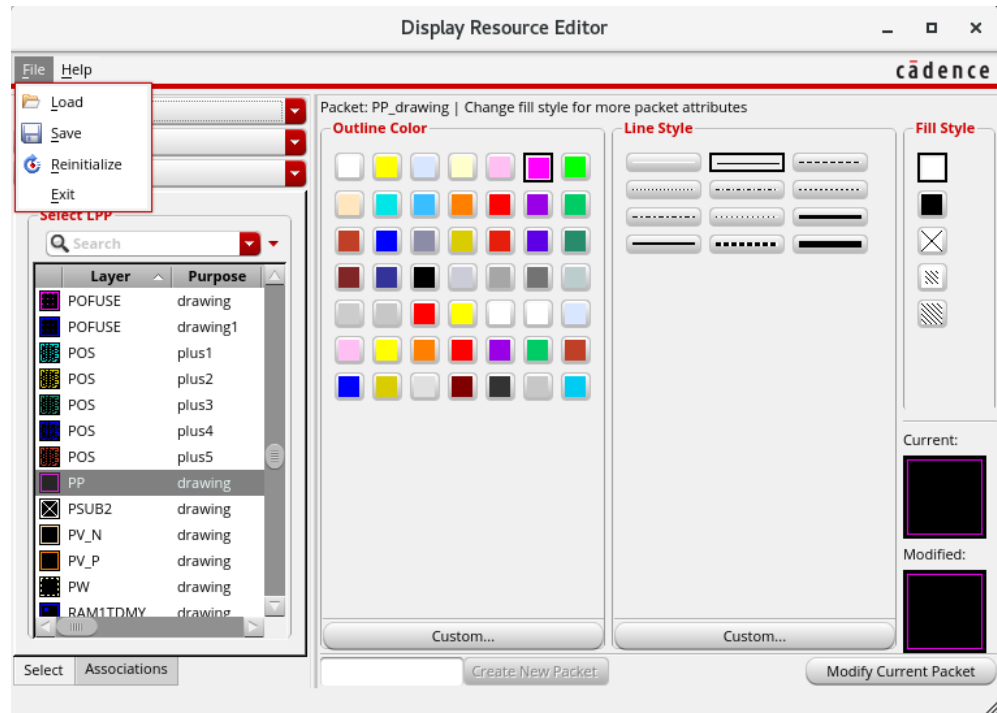
- b. To open the Display Resource Editor, click “Shift + Mouse Left Click” on any layer within the layer palette window



- c. Here are some of the things that you can do with the Display Resource Editor:

- i. View the display properties for layers
- ii. Edit the display properties for layers
- iii. Create new display packets

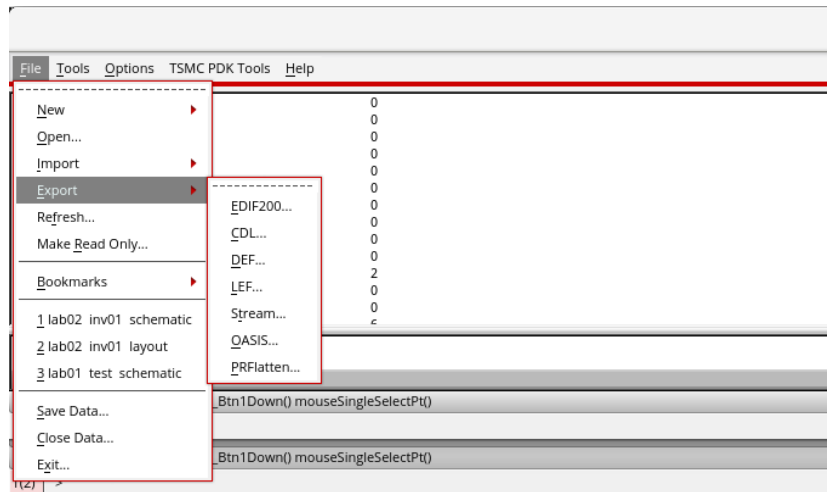
- iv. Merge display.drf files
- d. To Load a new display file , select File > Load > then browse to display file



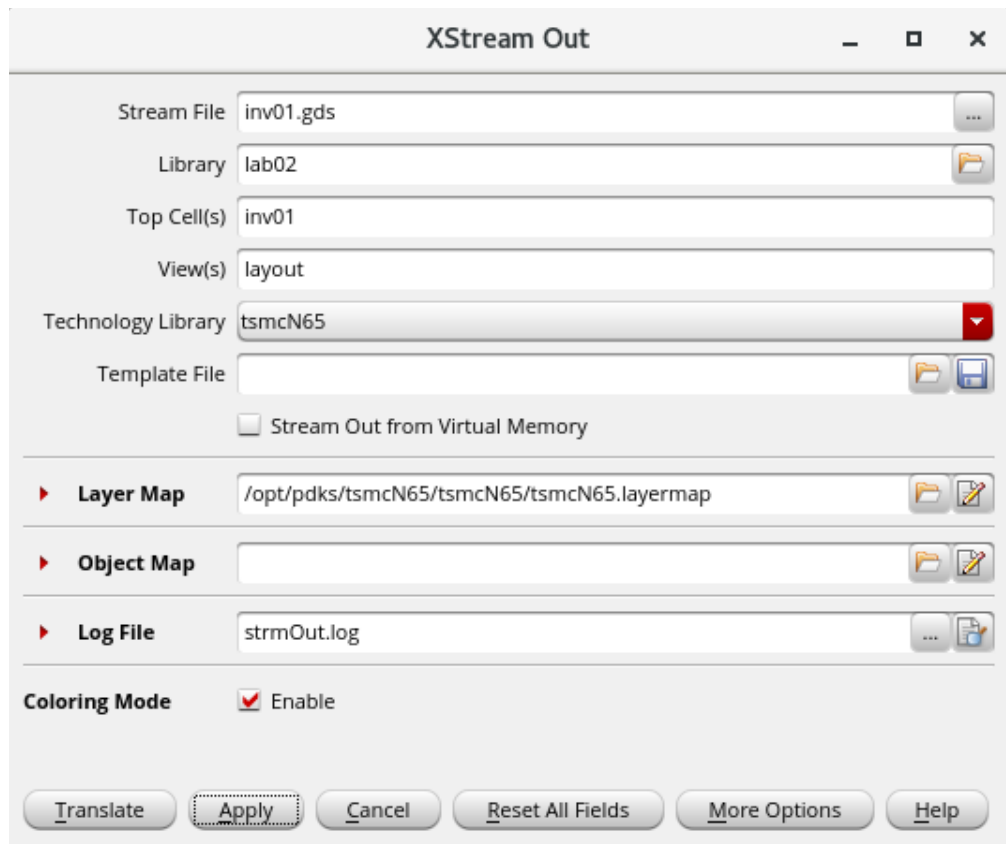
Part3: Exporting, importing and updating GDS using layer mapping

1. Exporting a GDS from Virtuoso

- a. To export a GDS, goto Virtuoso CIW > File > Export > Stream

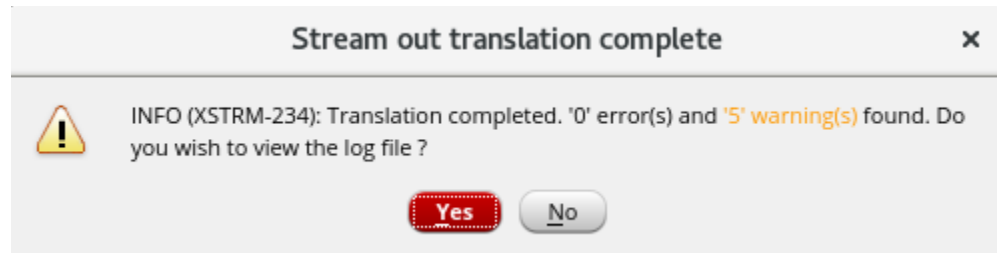


- b. Enter the library name, cell name, and layout view of part 1 layout design. And enter the technology layer map file path.



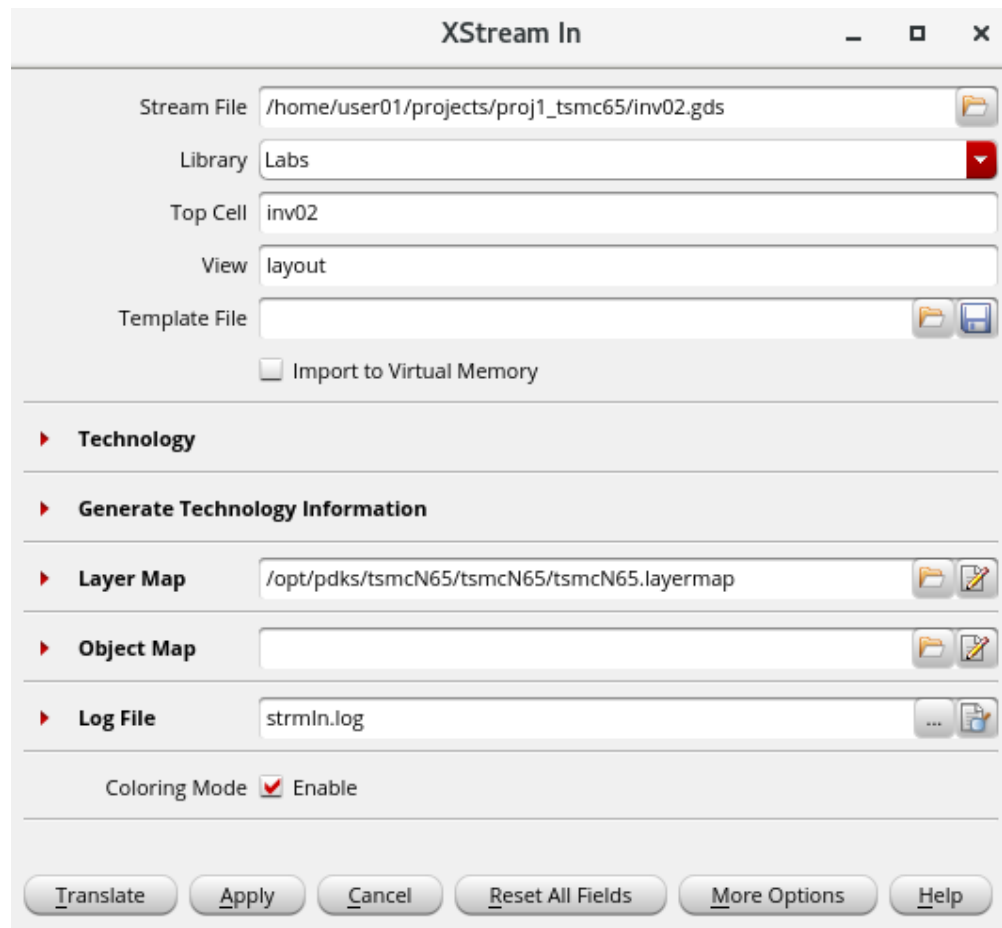
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- c. Translate and wait for the below popup window which should show 0 errors and this means a successful translation. However, you should show the log file and read the warning messages to make sure that they are not harmful by any means to the GDS export process.



2. Importing a GDS to Virtuoso

- a. Goto CIW > File > Import > Stream
- b. Browse to the GDS file to be imported, enter the destination library name, cell name (make it "inv02"), and the technology layer map file path
Note: Sometimes we may change the destination view name to be "layout_modified" for instance. But leave it as it is "layout" for now.



- c. Translate and make sure that translation was successful

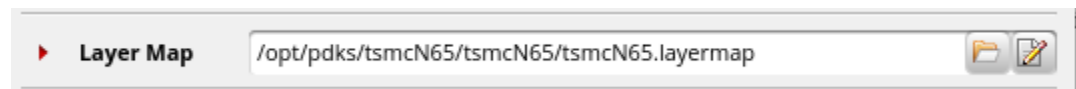
3. Layer map file

- a. A layer map file in Virtuoso is a text file that contains a mapping between the layer numbers in a GDS file and the layer names in the layout editor of any EDA vendor. The layer map file is used by Virtuoso/Custom-Compiler to convert between the two different layer identification systems. The layer map file is typically included in the Process Design Kit (PDK) provided by the FAB. It includes all the layers supported by the FAB for a specific technology node. The layer map file is a simple text file that consists of 4 columns: the layer name, the layer purpose, the stream layer number, and the stream layer purpose number.

- i. For example, the following line in a layer map file would map GDS layer number "1;0" to a Virtuoso layer name "DNW;drawing".

#Layer Name	Layer Purpose	Layer Stream Number	Datatype Stream Number
ref	drawing	0	0
DNW	drawing	1	0
PW	drawing	2	0
NW	drawing	3	0
PM	drawing	5	0
PM	drawing1	5	1
PM	drawing2	5	2
OD	drawing	6	0
OD	dummy	6	1
OD	drawing2	6	2
OD	drain	6	3
OD	pin	6	6
OD	test1	6	11
OD	test2	6	12
OD	test3	6	13
PDIFF	drawing	7	0
NDIFF	drawing	8	0
NT_N	drawing	11	0
NT_N	Ncap_NTN	11	20
VTL_N	drawing	12	0
VTL_P	drawing	13	0
OD_33	drawing	15	0

- b. The layer map file is used by Virtuoso when you import a GDS file. It is used to convert the GDS layer numbers to layer names. This allows you to view and modify the layout imported from the GDS file in Virtuoso.
- c. The layer map file is located at the following path



Lab Report

Prepare a PDF document in which you should include all the info/data required in the following tasks. Explicit attachments should not be part of the PDF document, instead they should be attached to the Lab assignment.

Task1

Report your progress in part 1 by reporting the following.

1. Add screenshots for the following
 - a. The schematic view showing the sizing
 - b. The first layout before running DRC and LVS
 - c. The layout after passing LVS and after the DRC final state.
 - d. The final DRC and LVS final reports.
2. Attach the LVS and DRC final reports which can be found under the following paths to the Lab assignment on Canvas
/home/<user-name>/verification/proj1_tsmc65/tsmcN65/drc/<cellname>.drc.summary
/home/<user-name>/verification/proj1_tsmc65/tsmcN65/lvs/<cellname>.lvs.report

Task2

Link between the layers used in the design of part 1, masks, and the fabrication steps and report the following.

Note: you may need to use what you learned in part 2 and you may need to check [the design rules document](#) P. 37.

1. The list of used layers and the fabrication steps based on the logical operation between the layers.
2. It would be more illustrative to use a table and to provide rectangular structures showing the fabrication steps on cross sectional views if possible.

Task3

Migrate “inv01” layout design which was based on LVT MOS devices to a new inverter “inv03” consisting of HVT MOS devices. You should make use of what you learned in part 3 to make it as fast as possible without the need to redesign the cell from scratch. And for the schematic, just copy it from “inv01” and edit the devices to be “pch_hvt” and “nch_hvt”.

Note: You may need to copy the layer map file and have a local one to modify and use.

After doing so please report the following.

1. Add screenshots for the following
 - a. The schematic view
 - b. The layout view
 - c. The modified layer map file highlighting the modification

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2. Attach the LVS and DRC final reports which can be found under the following paths to the Lab assignment on Canvas
/home/<user-name>/verification/proj1_tsmc65/tsmcN65/drc/<cellname>.drc.summary
/home/<user-name>/verification/proj1_tsmc65/tsmcN65/lvs/<cellname>.lvs.report
3. Attach the modified layer map file.